```r
library(tidyr)
library(dplyr)
library(ggplot2)
```

1 GLM for Binary Data

This uses the snoring levels as predictor to occurrence of heart attack. Refer to pg 118, Agresti.

```r
scoring <- c(0,2,4,5)

snoring<-rep(scoring,c(1379, 638, 213, 254))
y<-rep(rep(c(1,0),4),c(24,1355,35,603,21,192,30,224))
snoredata <- data.frame(y,snoring)
# snoredata$y <- factor(snoredata$y, levels = c(1,0), ordered = TRUE)
# snoredata$snoring <- factor(snoredata$snoring,
#                              levels = c(0,2,4,5), ordered = TRUE)

head(snoredata)
```

```
|    y snoring
|  1 1       0
|  2 1       0
|  3 1       0
|  4 1       0
|  5 1       0
|  6 1       0
```

Here is the contingency table:

```r
groupsnore <- data.frame(count=c(1379,638,213,254),
                         snore=scoring,
                         heart.disease = c(24,35,21,30))

groupsnore
```

```
|    count snore heart.disease
|  1  1379     0            24
|  2   638     2            35
|  3   213     4            21
|  4   254     5            30
```

```r
contingency_tab <- snoredata %>%
  group_by(snoring) %>%
  count(y) %>%
  pivot_wider(names_from = y, values_from = n) %>%
  mutate(totals = `1` + `0`)

contingency_tab
```

```
|  # A tibble: 4 x 4
|  # Groups:   snoring [4]
|    snoring   `0`   `1` totals
|      <dbl> <int> <int>  <int>
|  1       0  1355    24   1379
|  2       2   603    35    638
|  3       4   192    21    213
|  4       5   224    30    254
```

1.1 Linear Probability Model

For binary response variable, the regression model

$$\pi(\mathbf{x}) = \alpha + \beta_1 x_1 + \cdots + \beta_\rho x_\rho$$

is called a *linear probability model*. With independent observations, it is a GLM with binomial random component and identity link function.

(-): Probabilities fall between 0 and 1, but linear functions take values over the entire real line. The model can have $\pi(\mathbf{x}) > 1$ for some $\mathbf{x}$ values.

(+): its simple interpretation: β_j *is the change in* $\pi(\mathbf{x})$ for a one-unit increase in x_j

Since $var(Y) = \pi\left(\mathbf{x}\left[1 - \pi(\mathbf{x})\right]\right)$, the variance depends on $\mathbf{x}$ through its influence on $\pi(\mathbf{x})$. The constant variance condition that makes OLS estimators optimal is not satisfied, so the ML estimator is more efficient that least squares.

[Google AI] While glm with `family=binomial` is typically used for logistic regression (which uses a "logit" link), this uses `link="identity"`. This forces the model to treat the relationship between the predictors and the probability of the outcome as a straight line rather than a curve.

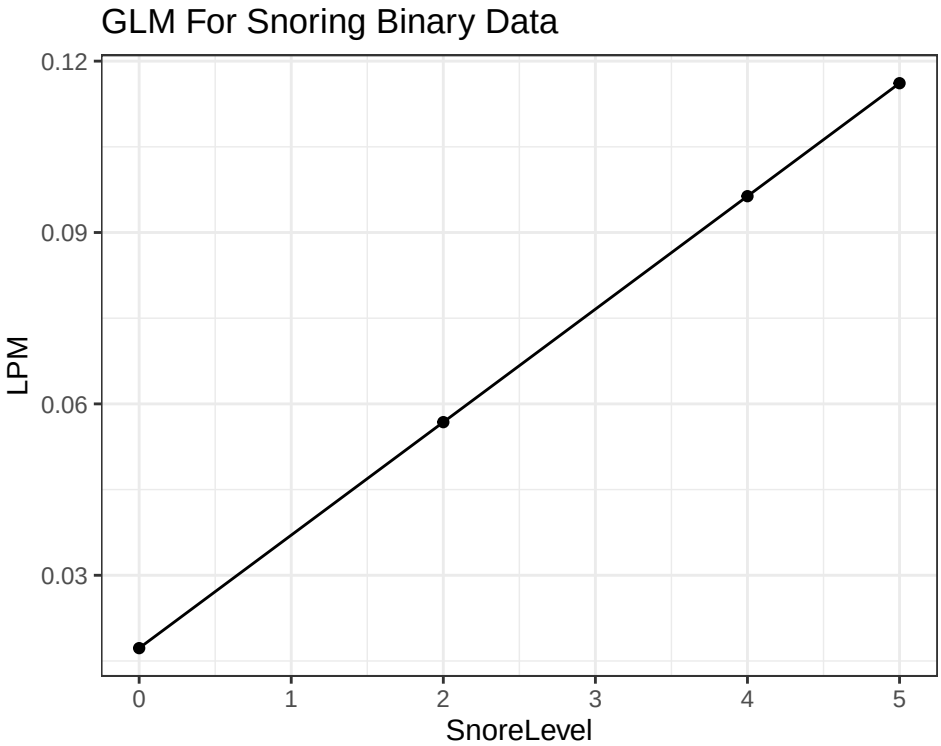
```r
fit <- glm(y~snoring,data=snoredata,family=binomial(link="identity"))
fit
```

```
|
|  Call:  glm(formula = y ~ snoring, family = binomial(link = "identity"),
|      data = snoredata)
|
|  Coefficients:
|  (Intercept)      snoring
|       0.0172       0.0198
|
|  Degrees of Freedom: 2483 Total (i.e. Null);  2482 Residual
|  Null Deviance:       901
|  Residual Deviance: 835   AIC: 839
```

For nonsnorers (`x=0`), the estimated proportion of subjects having heart disease is 0.01725. The estimated probability of heart disease is about 0.01725 for nonsnorers and it increases 2 × 0.01978=0.03956 for occasional snorers (`snoring = 2`), 4 × 0.01978=0.07911 for those who snore nearly every night (`snoring = 4`), and 5 × 0.01978=0.09889 for those who always snore (`snoring = 5`)

```r
new_obs <- data.frame(snoring = c(0, 2, 4, 5))
LPM_predictions <- predict(fit, newdata = new_obs, type = "response")
```

```r
data.frame(`SnoreLevel` = new_obs$snoring, LPM = LPM_predictions) %>%
  ggplot(aes(x = `SnoreLevel`, y = LPM)) +
  geom_point() +
  geom_line() +
  labs(title = "GLM For Snoring Binary Data") +
  theme_bw()
```



[Agresti] ML estimates and se are the same if we use a data file of 2484 binary observations (as shown above) or if we enter the four binomial totals of "yes" and "no" responses using `contingency_tab` (as shown below). Note: Set the first column as the success occurrence so that we get the same values as above.

```r
glm(cbind(contingency_tab$`1`, contingency_tab$`0`) ~ contingency_tab$snoring,
    family=binomial(link="identity"))
```

```
|
|  Call:  glm(formula = cbind(contingency_tab$‘1‘, contingency_tab$‘0‘) ~
|      contingency_tab$snoring, family = binomial(link = "identity"))
|
|  Coefficients:
|              (Intercept)  contingency_tab$snoring
|                   0.0172                   0.0198
|
|  Degrees of Freedom: 3 Total (i.e. Null);  2 Residual
|  Null Deviance:       65.9
|  Residual Deviance: 0.0692    AIC: 24.3
```

See that it reverses when columns reverse. It just differs in interpretation.

```r
altfit <- glm(
  cbind(contingency_tab$`0`, contingency_tab$`1`) ~ contingency_tab$snoring,
  family=binomial(link="identity"))
altfit
```

```
|
|  Call:  glm(formula = cbind(contingency_tab$`0`, contingency_tab$`1`) ~
|      contingency_tab$snoring, family = binomial(link = "identity"))
|
|  Coefficients:
|              (Intercept)  contingency_tab$snoring
|                   0.9828                  -0.0198
|
|  Degrees of Freedom: 3 Total (i.e. Null);  2 Residual
|  Null Deviance:       65.9
|  Residual Deviance: 0.0692    AIC: 24.3
```

The estimated probability of not having heart disease is about 0.98275 for nonsnorers and it decreases $-1\times$ 2 $\times$ -0.01978=0.03956 for occasional snorers (`snoring = 2`), $-1\times$ 4 $\times$ -0.01978=0.07911 for those who snore nearly every night (`snoring = 4`), and $-1\times$ 5 $\times$ -0.01978=0.09889 for those who always snore (`snoring = 5`)

Notes:

- you get completely different model when you treat `snoring` as categorical rather than continuous. It depends how you want to model. Treating it as continuous is parsimonious.

1.2 Logistic regression

Usually binary data result from a nonlinear relationship between $\pi(x)$ and x. A fixed change in x often has less impact when $\pi(x)$ is near 0 or 1 than when $\pi(x)$ is near 0.50.

In practice, nonlinear relationships between $\pi(x)$ and x are often monotonic, with $\pi(x)$ increasing continuously or $\pi(x)$ decreasing continuously as x increases. The most important curve with this shape has the model formula,

This is a logistic regression model. As x increases, $\pi(x)$ increases when $\beta > 0$ and decreases when $\beta < 0$. Logistic regression is a GLM, the link is the log odds transformation — the logit. Logistic regression models are GLMs with binomial random component and logit link function.

$$\begin{aligned}
\pi(x) &= \frac{\exp(\alpha+\beta x)}{1+\exp(\alpha+\beta x)} \\
\pi(x) + \pi(x)\exp(\alpha+\beta x) &= \exp(\alpha+\beta x) \\
\pi(x) &= \exp(\alpha+\beta x) - \pi(x)\exp(\alpha+\beta x) \\
\frac{\pi(x)}{1-\pi(x)} &= \exp(\alpha+\beta x) \\
\log\frac{\pi(x)}{1-\pi(x)} &= \alpha+\beta x
\end{aligned}$$

Whereas $\pi(x)$ must fall in the $(0,1)$ range, the logit can be any real number. The real numbers are also the range for linear predictors that form the systematic component of a GLM. So, this model does not have the structural problem that LPM has.

```
|   function (x, ..., wt = NULL, sort = FALSE, name = NULL)
|   {
|       UseMethod("count")
|   }
|   <bytecode: 0x000001c07ac923c0>
|   <environment: namespace:dplyr>
```

```r
logitfit <- glm(
  cbind(contingency_tab$`1`, contingency_tab$`0`) ~ contingency_tab$snoring,
  family=binomial(link="logit"))
logitfit
```

```
|
|   Call:  glm(formula = cbind(contingency_tab$`1`, contingency_tab$`0`) ~
|       contingency_tab$snoring, family = binomial(link = "logit"))
|
|   Coefficients:
|               (Intercept)  contingency_tab$snoring
|                    -3.866                    0.397
|
|   Degrees of Freedom: 3 Total (i.e. Null);  2 Residual
|   Null Deviance:       65.9
|   Residual Deviance: 2.81  AIC: 27.1
```

```r
# same
# glm(contingency_tab$`1`/contingency_tab$totals~contingency_tab$snoring,
# family=binomial(link="logit"),weights=contingency_tab$totals)
```

```r
fit2 <- glm(heart.disease/count~snore,data=groupsnore,family=binomial(link="logit"),weig
# summary(fit) # display results
# confint(fit,level = 0.99) # 99% CI for the coefficients
# exp(coef(fit)) # exponentiated coefficients
# exp(confint(fit)) # 99% CI for exponentiated coefficients
# predict(fit, type="response") # predicted values
# residuals(fit, type="deviance") # residuals
```

2 Loglinear Models for contingency tables

- Loglinear models are GLMs using the log link function with a Poisson response.
- Common use: modeling for contingency tables.
- The models specify the joint distribution among the categorical variables that are cross-classified to form the table.
- They are used to analyze association and interaction patterns among the variables.
- the models specify how the expected cell counts depend on levels of the categorical variables for that ceel as well as associations and interactions among those variables
- Loglinear models are of use primarily whan at least two variables are response variables. With a single categorical response, it is more natural to use logistic regression models.
- When one variable is treated as a response and the others as explanatory variables, logistic models for that response variable are equivalent to certain loglinear models. (Section 9.5)

2.1 Loglinear models for 2-way tables

- Let there be $I \times J$ contingency table cross-classifying a multinomial sample of n subjects on two categorical responses.
- For cell probabilities π_{ij} the expected frequencies are $\mu_{ij} = n\pi_{ij}$ (Notice it's just the proportion)
- Loglinear model formulas use μ_{ij} rather than π_{ij}

2.1.1 Independence model for 2-way table

Under statistical independence,

$$\mu_{ij} = \mu\alpha_i\beta_j$$

For multinomial sampling for instance, $\mu_{ij} = n\pi_{i+}\pi_{+j}$.

Denote the row variable by X and the column variable by Y. The formula expressing independence is multiplicative. Thus, $\log \mu_{ij}$ has additive form

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y$$

for a row effect λ_i^X and a column effect λ_j^Y. This is the loglinear model of independence.

Identifiability requires constraint such as $\sum_i \lambda_i^X = \sum_j \lambda_j^Y = 0$

GoF Tests: χ^2, G^2

2.1.2 Interpretation

See additional reference here

[AI Supplement on understanding row and column effects]

To understand row and column effects, it helps to stop looking at individual cells for a moment and look only at the marginal totals (the sums at the ends of the rows and columns). 1. The Row Effect (λ_i^X) - The row effect tells you how much "weight" a specific row carries in your total sample, regardless of what's happening in the columns. - What it represents: It accounts for the fact that some categories of your row variable are naturally more common than others. - The "Goal": The model uses to ensure that the predicted sum of a row matches the actual total count observed for that row. - Intuition: If Row 1 is "Doctors" and Row 2 is "Astronauts," and you have 1,000 doctors but only 10 astronauts, will be much larger than - It tells the model, "Hey, expect a lot more people in the first row just because there are more doctors in our study". 2. The Column Effect (λ_i^X) - The column effect does exactly the same thing, but for the columns. - What it represents: It accounts for the popularity or frequency of the column categories. - The "Goal": It ensures the predicted sum of each column matches the actual total count observed for that column. - Intuition: If Column 1 is "Likes Pizza" and Column 2 is "Likes Anchovies," and almost everyone likes pizza but few like anchovies, will be high and will be low.

Listed references: - https://education.illinois.edu/docs/default-source/carolyn-anderson/edpsy589/lectures/6-loglinear/6_loglinear_models_beamer_post.pdf - https://online.stat.psu.edu/stat504/Lesson10#:~:text=Recall%2C%20that%20the%20independence%20can,%2Dparametriza

3 Logistic Regression

3.1 Parameter interpretation

For a binary response variable `Y` and an explanatory variable `X`, let $\pi(x) = P(Y = 1 \mid X = x)1 - P(Y = 0 \mid X = x)$. The logistic regression model is

$$\pi(x) = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)} \tag{3.1}$$

$$\begin{aligned}
\pi(x) &= \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)} \\
&\longleftrightarrow \\
\text{logit}[\pi(x)] &= \log \frac{\pi(x)}{1 - \pi(x)} \\
&= \log \exp(\alpha + \beta x) \\
&= \alpha + \beta x
\end{aligned} \tag{3.2}$$

$$\begin{aligned}
\frac{\pi(x)}{1 - \pi(x)} &= \frac{\frac{\exp(\alpha+\beta x)}{1+\exp(\alpha+\beta x)}}{1 - \frac{\exp(\alpha+\beta x)}{1+\exp(\alpha+\beta x)}} \\
&= \frac{\frac{\exp(\alpha+\beta x)}{1+\exp(\alpha+\beta x)}}{\frac{1}{1+\exp(\alpha+\beta x)}} \\
&= \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)} \times \frac{1 + \exp(\alpha + \beta x)}{1} \\
&= \exp(\alpha + \beta x)
\end{aligned}$$

Interpreting β in Equation (3.2):

- its sign determines where $\pi(x)$ is increasing or decreasing as x increases
- The rate of climb or descent increases as $\mid \beta \mid$ increases
- As $\beta \to 0$, the cure flattens to a horizontal straight
- When $\beta = 0$, Y is independent of X
- For quantitative x with $\beta > 0$, the curve for $\pi(x)$ has the shape of the cdf of the logistic distribution. Since the logistic density is symmetric, $\pi(x)$ approaches 1 at the same rate that it approaches 0

Exponentiating both sides of Equation (3.2),

$$\begin{aligned}
\exp(\text{logit}[\pi(x)]) = \exp\left(\log \frac{\pi(x)}{1 - \pi(x)}\right) &= \exp(\alpha + \beta x) \\
\longleftrightarrow \frac{\pi(x)}{1 - \pi(x)} &= \exp(\alpha)\exp(\beta x) \qquad \text{odds at X = x}
\end{aligned} \tag{3.3}$$

What happens when x increases by 1 unit? The odds multiply by $\exp(\beta)$ for every 1-unit increase in x.

$$\begin{aligned}
\frac{\pi(x+1)}{1-\pi(x+1)} &= \exp(\alpha)\exp(\beta(x+1)) \\
&= \exp(\alpha)\exp(\beta x+\beta) \\
&= \exp(\alpha)\exp(\beta x)\exp(\beta) \qquad \text{odds at X = x+1} \\
&= \frac{\pi(x)}{1-\pi(x)}\exp(\beta) \qquad \text{substitute odds at X = x} \\
&= \frac{\pi(x)}{1-\pi(x)}\exp(\beta) \\
\frac{\frac{\pi(x+1)}{1-\pi(x+1)}}{\frac{\pi(x)}{1-\pi(x)}} &= \exp(\beta) \qquad \exp(\beta) \text{ is an odds ratio}
\end{aligned} \tag{3.4}$$

In other words, $\exp(\beta)$ is an odds ratio — the odds at $X = x+1$ divided by the odds at $X = x$

The logistic regression function in Equation (3.5) implies that the rate of change in $\pi(x)$ per unit change in x varies. A straight line'

$$\frac{\partial\pi(x)}{\partial x} = \frac{\partial}{\partial x}\frac{\exp(\alpha+\beta x)}{1+\exp(\alpha+\beta x)} \tag{3.5}$$

$$= \frac{(1+\exp(\alpha+\beta x))\frac{\partial}{\partial x}\exp(\alpha+\beta x) - \exp(\alpha+\beta x)\frac{\partial}{\partial x}(1+\exp(\alpha+\beta x))}{(1+\exp(\alpha+\beta x))^2} \tag{3.6}$$

$$= \frac{(1+\exp(\alpha+\beta x))\beta\exp(\alpha+\beta x) - \exp(\alpha+\beta x)\beta\exp(\alpha+\beta x)}{(1+\exp(\alpha+\beta x))^2} \tag{3.7}$$

$$= \frac{\beta\exp(\alpha+\beta x) + \beta\exp(\alpha+\beta x)^2 - \beta\exp(\alpha+\beta x)^2}{(1+\exp(\alpha+\beta x))^2} \tag{3.8}$$

$$= \beta\frac{\exp(\alpha+\beta x)}{1+\exp(\alpha+\beta x)}\frac{1}{1+\exp(\alpha+\beta x)} \tag{3.9}$$

$$= \beta\pi(x)(1-\pi(x)) \tag{3.10}$$

4 Appendix

$$
\begin{aligned}
P(Y_i = 1|X_i) &= \beta_0 + \beta_1 X_i \\
P(Y_i = 0|X_i) &= 1 - P(Y_i = 1|X_i) = 1 - (\beta_0 + \beta_1 X_i) \\
Y_i &\sim Bernoulli(P(Y_i = 1|X_i)) \\
P(Y_i|X_i) &= 1 - P(Y_i = 1|X_i) = 1 - (\beta_0 + \beta_1 X_i) \\
P(Y_i = y_i|X_i) &= P(Y_i = 1|X_i)^{y_i} \left(1 - P(Y_i = 1|X_i)\right)^{1-y_i}
\end{aligned}
$$

$$
\begin{aligned}
P(Y_1 = y_1, \ldots, Y_n = y_n|X_1 \ldots X_n) &= \prod_{i=1}^{n} \left[P(Y_i = 1|X_i)^{y_i} \left(1 - P(Y_i = 1|X_i)\right)^{1-y_i}\right] \\
&= \log \left[\prod_{i=1}^{n} \left[P(Y_i = 1|X_i)^{y_i} \left(1 - P(Y_i = 1|X_i)\right)^{1-y_i}\right]\right] \\
&= \sum_{i=1}^{n} \log \left[P(Y_i = 1|X_i)^{y_i} \left(1 - P(Y_i = 1|X_i)\right)^{1-y_i}\right] \\
&= \sum_{i=1}^{n} y_i \log P(Y_i = 1|X_i) + (1 - y_i) \log \left(1 - P(Y_i = 1|X_i)\right) \\
&= \sum_{i=1}^{n} y_i \log(\beta_0 + \beta_1 X_i) + (1 - y_i) \log \left(1 - (\beta_0 + \beta_1 X_i)\right) \\
0 &= \sum_{i=1}^{n} y_i \frac{1}{\beta_0 + \beta_1 X_i} \frac{\partial X_i}{\partial \beta_1}
\end{aligned}
$$